

Statistical Inference

Statistical inference is a set of methods by which we try to draw a conclusion about the population based on information obtained from a sample.

Statistical inference: ... induce or infer the properties of a population [parameters] on the basis of known data relating to a sample [statistics]

(H. M. Blalock).

Statistical Inference

Only exceptionally do we know directly the characteristics of the population, we usually have to estimate them starting from the characteristics of the samples that have been extracted from the population.

Statistical inference aims at testing a relative hypothesis the characteristics of the population.

The procedure first involves the formulation of the hypothesis and then the evaluation of the probability of obtaining those results in the population if the starting hypothesis were true.

The logical path of inference takes place according to the following phases:

1. Extraction of a part of the population (sample)
2. Calculation of sample statistics, i.e. the values corresponding to the data contained in the sample (e.g. sample mean)
3. Estimation of parameters in the population based on the results provided by the sample (inference).

Questo percorso è essenziale per la costruzione dell'evidenza scientifica, in particolare quando studiamo fenomeni biologici, che sono affetti da elevata variabilità.

Population:

set that collects all possible observations, relating to a given variable or a given phenomenon. it can be finite (however very large) or infinite

Sample:

- finite collection of elements extracted from a population
- the purpose of the extraction is to obtain information on the population
- therefore the sample must be representative of the population from which it is extracted ('not spoiled')
- to correspond to these needs, the sample is identified with a random sampling.

The use of samples is necessary because:

- 1) the reference population constitutes an infinite universe (all subjects affected by lung cancer in the history of humanity, including the future) that cannot be known in a way exhaustive.
- 2) the population represents a finite universe which, however, due to cost and time problems, cannot be explored in its entirety.

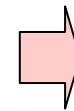
In both cases, a sample that is as representative as possible of the population is used (equal to the population with respect to the characteristics considered salient for the research), in order to generalize the results obtained in the sample to the entire reference population (inference).

Types of sampling:

a simple random sampling all individuals in the population are equally likely to be included in the sample.

- individuals in the population = "sampling unit"
- study population = "target population"
- actually sampled population (net of the effect of selection factors) = "study population" or sampling basis

If sampling basis $\neq$ target population



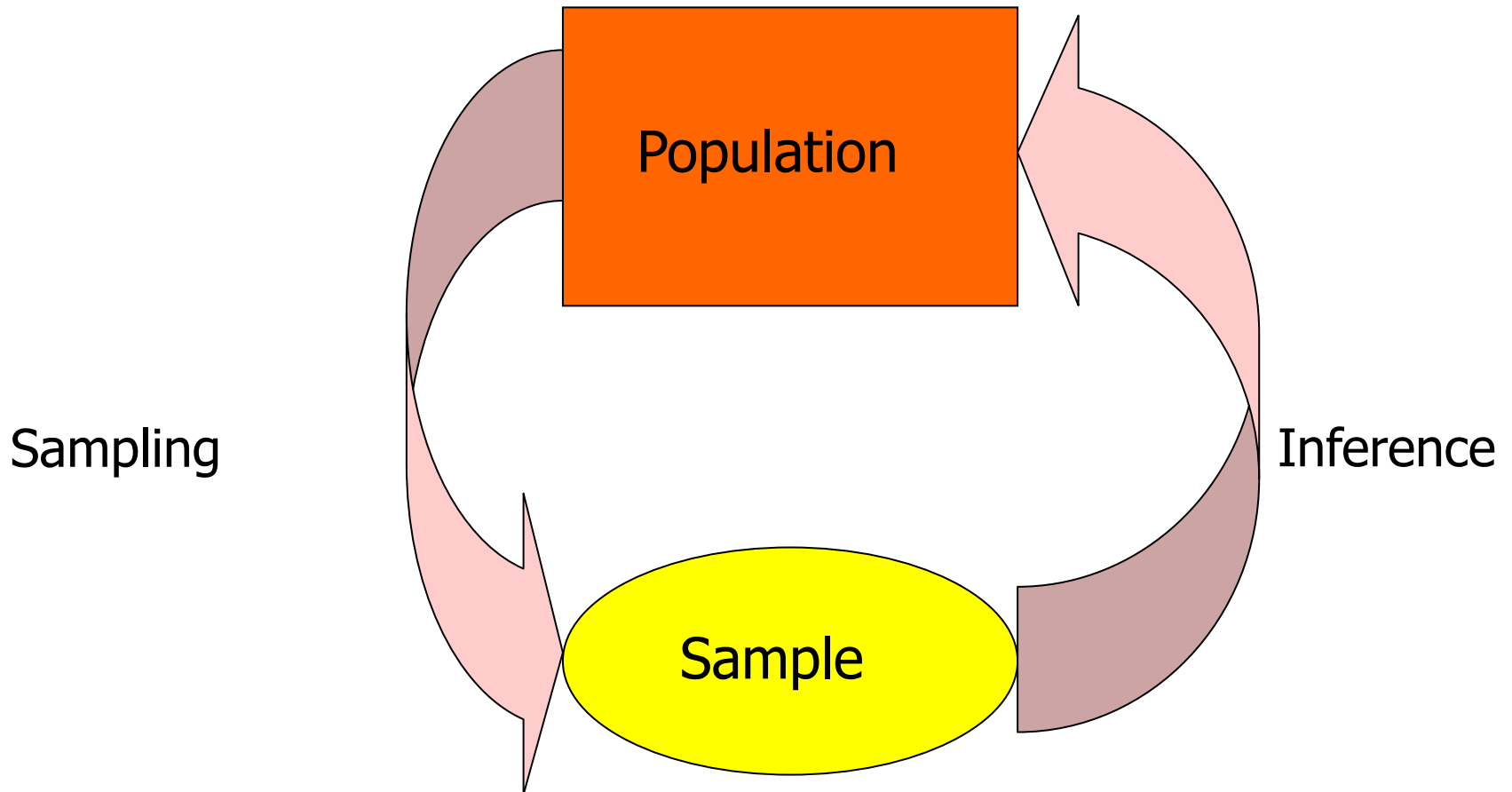
BIAS

d i s t o r t e d
sample

In non-random sampling not all units can be sampled. There are elements that have a different probability to become part of the sample (when for example we choose i subjects based on personal knowledge ...).

It is not possible to make inference on non-random samples.

Statistical inference and sampling



Sampling and inference are two symmetrical processes.

Some reminders

A numeric variable that can take on several values is called a random variable. The random attribute refers to the fact that it is generated by an experiment (or mechanism, natural phenomenon etc.) of which we are unable to predict the outcome with certainty.

Each of the results of a random variable is associated with a certain probability.

The function that associates a probability to each value of the variable is called "probability distribution".

The total area subtended by a probability distribution is equal to 1.

The probability distributions of many variables can be determined on a theoretical basis: "theoretical probability distributions"



Theoretical distributions or probability laws "intervene in inferential statistics as a tool to represent an observed distribution by means of a mathematical model that depends on a limited number of parameters".

A model enters an inference process in two ways:

- some problems of statistical inference to be solved require particular assumptions about the shape of the distribution that characterizes the population;
- other inference problems, on the other hand, while not requiring particular distributional assumptions on the population, are based on the characteristics of a theoretical distribution.

In both cases we need a mathematical model to study the random phenomenon (regulated by chance) from an inferential point of view.

Probability distributions:

Any feature that can be measured or categorized represents a variable.

If a probability is associated with each value that the variable can assume, intended as the relative frequency of the occurrence of each result x in numerous repeated experiments, then we speak of probability distribution.

The normal or GAUSSIAN distribution

It is the probability distribution that best represents many biological variables.

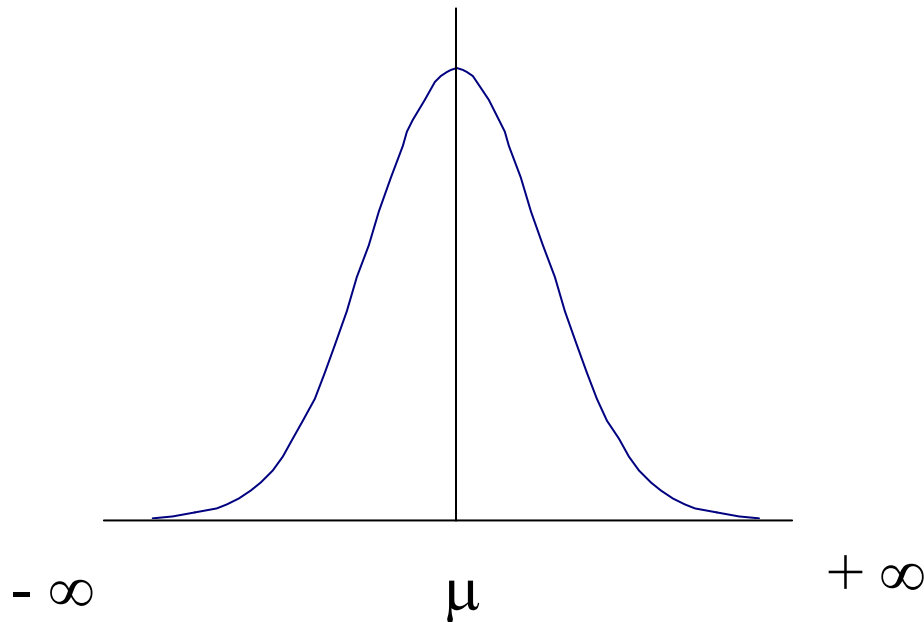
For example weight, blood pressure, cholesterol level in the blood are some of the variables that follow one normal distribution.

it applies well to sample statistics

The normal distribution formula is defined by the parameters mean (μ) and standard deviation (σ).

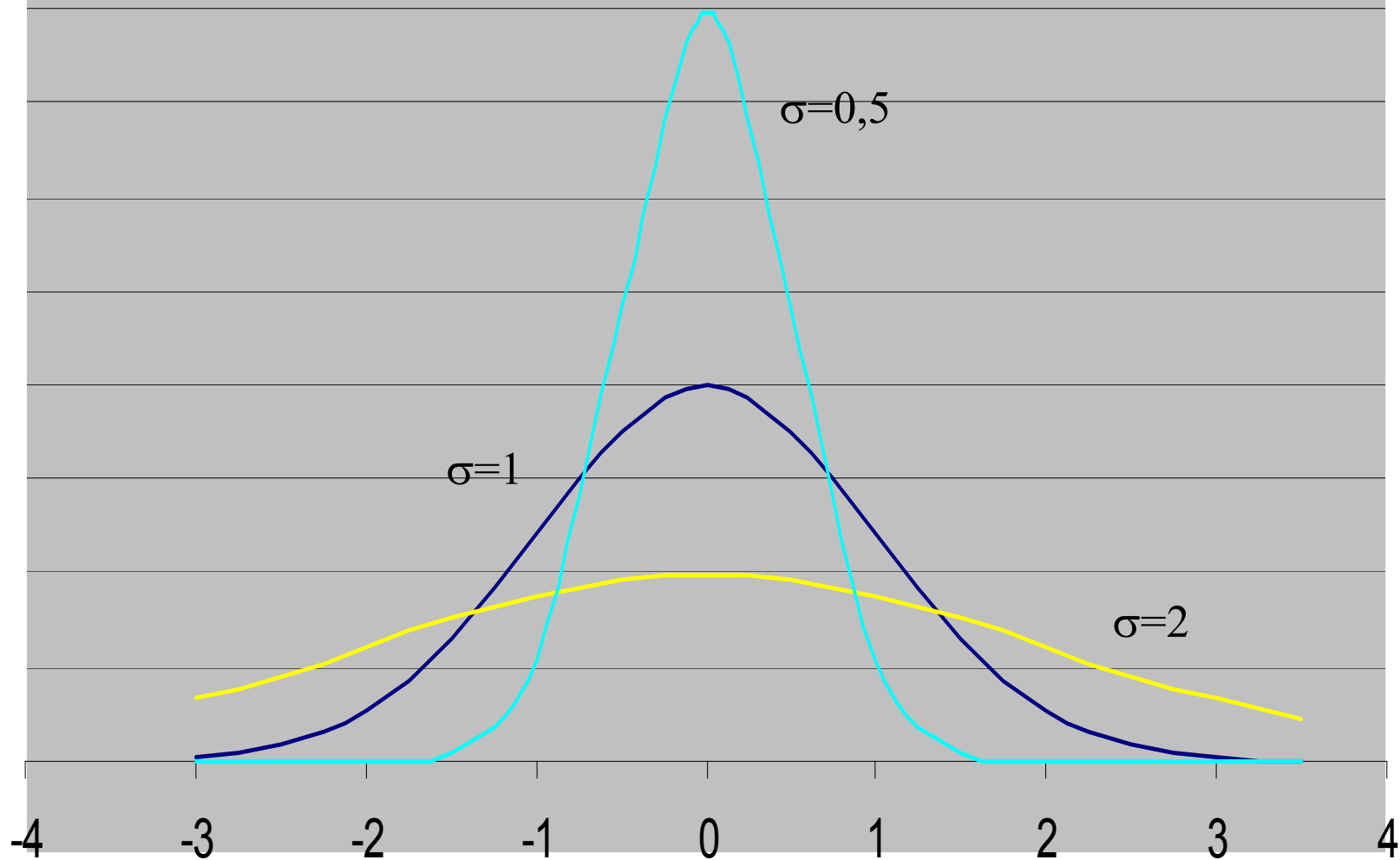
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} * \exp \frac{-(x - \mu)^2}{2\sigma^2}$$

The normal or GAUSSIAN distribution



- theoretical probability distribution concerning continuous variables with values between $(-\infty, +\infty)$
 - *depends on the parameters μ (mean) and σ (standard deviation); if $\mu = 0$ and $\sigma = 1$ it is called Standard Normal*
- it is symmetrical with respect to μ and changes concavity at the point $\mu \pm \sigma$; mode, mean and median coincide in μ .

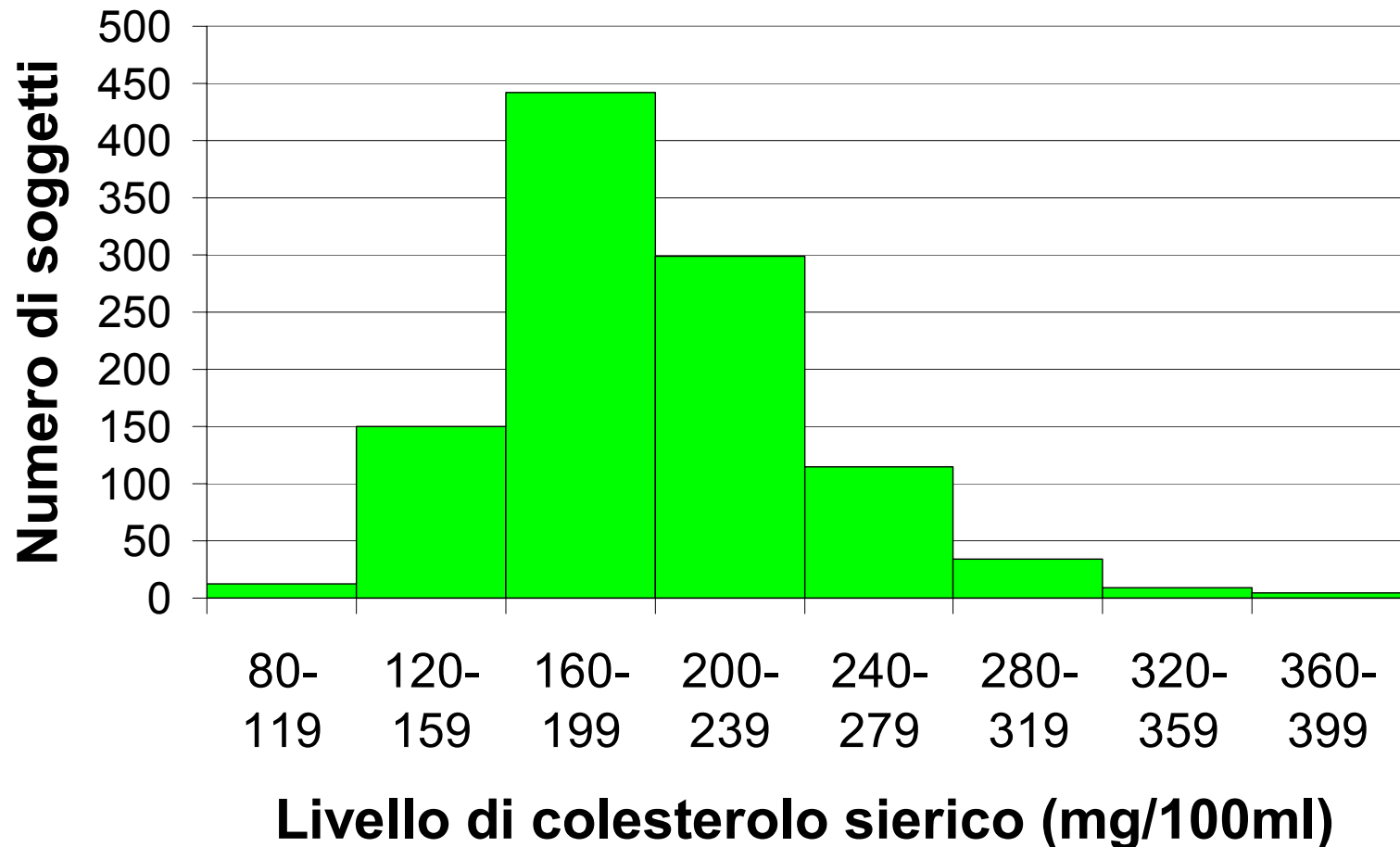
media $\mu=0$ e dev.std. σ



media μ e dev.std $\sigma=1$



Example of a variable that distributes normally: serum cholesterol levels in 1067 subjects of the male population of the United States between the ages of 25 and 34, 1976-1980.



What are the characteristics of this distribution and why is it so important?

The normal curve can be used to estimate the probabilities associated with variables that are “normally” distributed.

For example, in a population where cholesterol levels are normally distributed with mean $\mu = 220$ and standard deviation $\sigma = 21$, we might want to know the probability that a randomly chosen individual has a serum cholesterol value greater than 250 mg / 100ml.

Since the total area under the curve is equal to 1, we can estimate this probability by determining the proportion of the area under the curve lying to the right of $x = 250$.

To answer this question I can use statistical tables in which all the probabilities assigned to each value of this distribution with mean 0 and standard deviation 1 are tabulated.

Since a normal distribution can have an infinite number of values for its mean and standard deviation, a random variable X distributed as a Gaussian can be traced back to the Standard Normal which has mean 0 and standard deviation 1 ($\mu = 0$ and $\sigma^2 = 1$), through the transformation:

$$Z = \frac{x - \mu}{\sigma}$$

where:

x : value we are interested in

μ : mean in the population

σ : standard deviation in the population

Z : standardized normal deviation corresponding to the values given for (x , σ , μ).

With this procedure the Gaussian distribution (of population) of any parameters μ and σ^2 to the Standard Normal for which there are numerical probability tables.

For each value of the variable that I am measuring I can calculate the portion of the area that that value leaves to its right and this value corresponds to the probability that I have of observing that value.

For example, to calculate the probability of observing a subject with a cholesterol value greater than or equal to 250, I calculate the value of Z

$$Z = (250 - 220) / 21 = 1,43$$

And then I go to look on the statistical tables what is the probability that this value leaves to its right. $P(Z \geq 1.43) = 0,0076$

Therefore the value 250 has a probability of occurring equal to 7% in a population with mean 220 and standard deviation 21.

Properties of the standard normal distribution

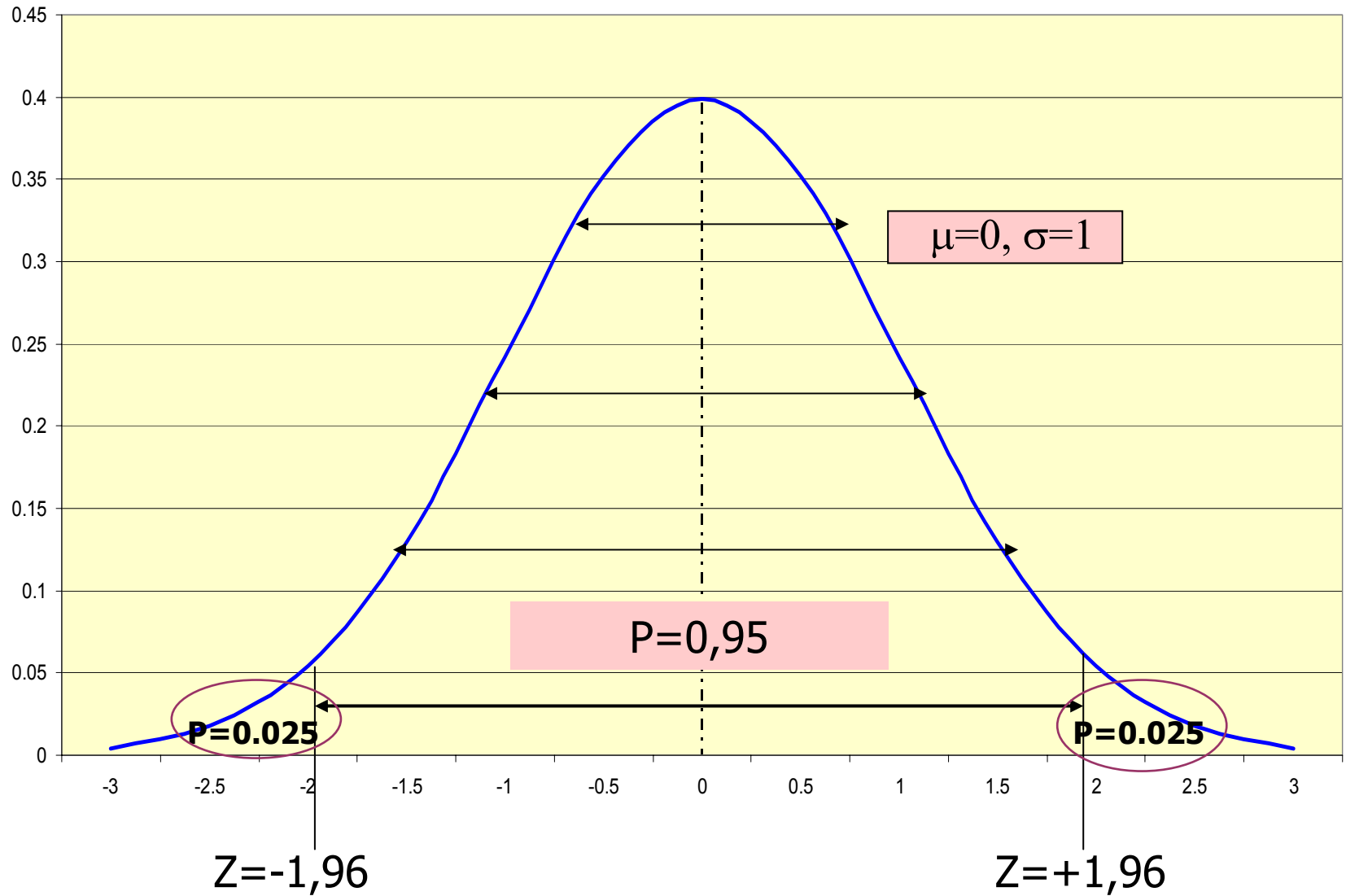
It is a distribution:

- Simmetric
- The total area under the curve represents the probability and will be equal to 1
- mean and median coincide
- it is described by only two parameters: mean (μ) and standard deviation (σ).

ATTENZION:

A mathematical property of the standard normal distribution is that exactly 95% of the values are between the mean ± 1.96 standard deviations

Standardized normal probability distribution

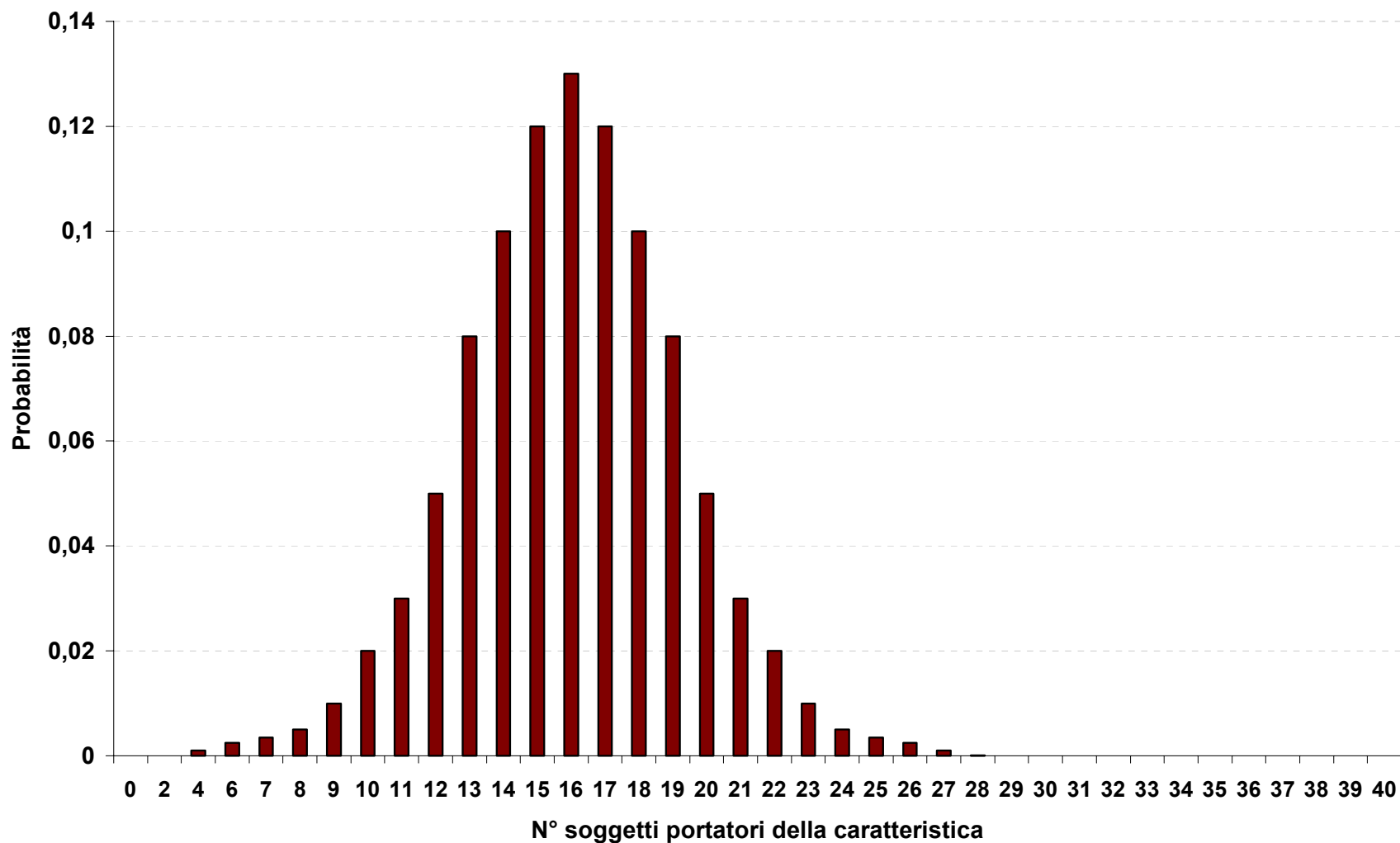


Estimating the proportion through sampling:

meaning and use of BINOMIAL DISTRIBUTION

- Let's imagine we have an infinite population of subjects in which a given feature is present in 40% of cases.
- Suppose we randomly extract numerous from that population samples of 40 subjects each.
- Not all samples will have the same number of subjects with the characteristic date; although we can predict that the most of them have 16 (40%), there will be some which will have different compositions.
- With statistics we have the possibility to calculate a priori the probability of extracting each sample of 40 subjects from this population as a function of the number of individuals carrying the characteristic under study.

Probability of randomly obtaining all possible samples of 40 subjects from a population in which 40% of the individuals have a given characteristic



We can note that (by doing a random extraction) the easiest sample to obtain is precisely the one in which 16 subjects present the characteristic under study ($P = 0.13$), while the most difficult to obtain is the one in which all 40 present it ($P = 0.00000000000001$).

The distribution described is the BINOMIAL

$$P(X = x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

where

X is the binomial random variable (it assumes only 2 mutually exclusive values: presence or absence of an attribute)

n is the number of the extracted sample

p is the probability of success (between 0 and 1)

The binomial distribution expresses the probability that a certain combination of subjects (relative to the presence or absence of the character) has to occur

The easiest sample to obtain is the one with the same characteristics as the population, while those increasingly distant have less and less probabilities of being extracted: it is therefore possible to identify samples whose probabilities are so low as to be considered negligible.

How far can we go so that the mistake we have decided to commit does not become so great as to invalidate our inferences?

By convention, it is believed that all phenomena whose probabilities of occurring are not likely to be considered implausible exceed, if added together (cumulative probability), 5% (or 1% if we are even more conservative)

Distribution queues (2.5% from each of the queues)

The set of these samples constitutes what is called a region of rejection or 5% significance; the others are part of the non-rejection region.

Inference on means

In estimating a population parameter, a point estimate is calculated on a representative sample (starting from the information taken from a sample, I obtain a single numerical value used as an estimate of the parameter of the entire population) and a range of values (interval estimate) in which the parameter lies with a certain degree of certainty.

This is one of the two ways of proceeding in statistical inference, as it is possible alternatively to proceed by hypothesis testing.

It is intuitive that the estimate is better, so much so:

- the greater the sample size n
- the smaller the variability between individuals

Definition of sample distribution of means:

If we repeat the random sampling operation a number of times large enough to extract from the population all possible samples of size n and for each we calculate the average:

$$\overline{X_1} \neq \overline{X_2} \neq \overline{X_3} \neq \dots \overline{X_i}$$

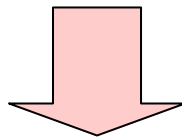
The sample distribution of the means is the distribution of the means of the samples extracted from the population.

Repeated samples from the same population yield different sample means

Each of these sample means constitutes an undistorted estimate of the parameter (population mean) but cannot be used as an estimate of the parameter alone, without taking into account the uncertainty caused by the sampling error.

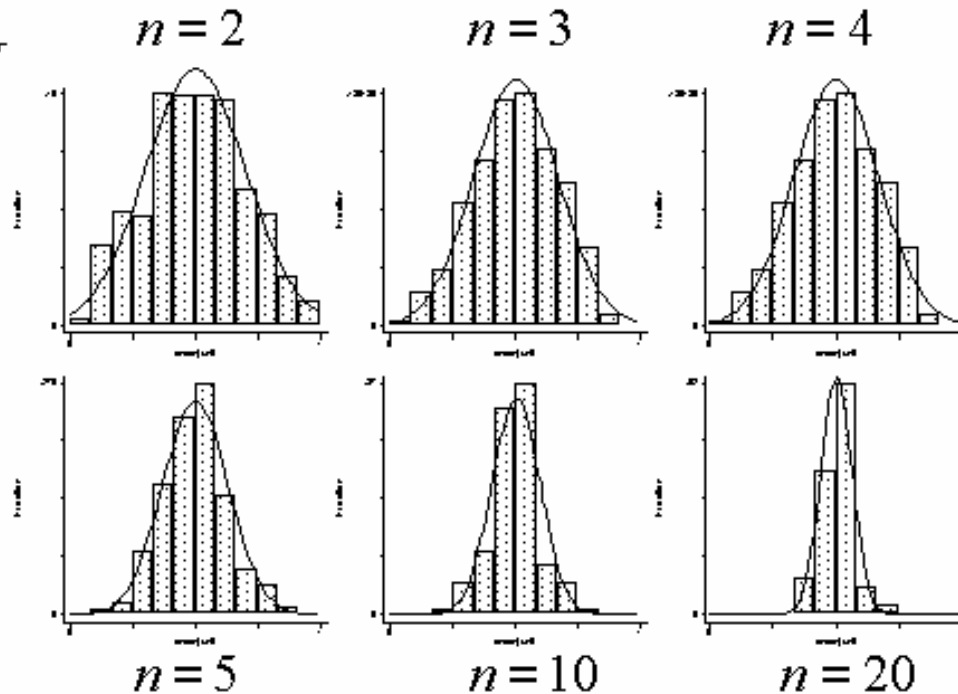
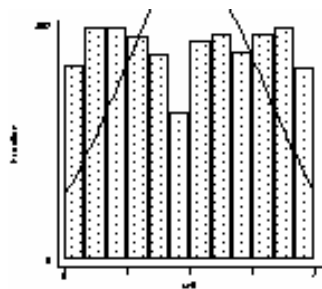
Rules:

- The mean of the sampling distribution of the means is equal to the population mean μ
- The variability of the distribution of the sample means is lower to the variability in the population. Larger samples will give one distribution with lower variability
- The d.s. of the sampling distribution is $\sigma / \sqrt{n}$ and this quantity is known as the Standard Error
- The shape of the sampling distribution is normal if n is sufficiently large (whatever the shape of the distribution population) - Central limit theorem



The probability distribution relevant for making inferences on the means is the Gaussian.

Population with uniform distribution (1000 observations)



As the sample size increases, the distribution of sample averages is concentrated around the population mean. The variance decreases as the sample size increases.

The **Standard Deviation (SD)** expresses the average dispersion of the observations of a sample.

The **Standard Error (ES)** is the SD of the means and expresses how close the average calculated on a sample is to the true mean. Therefore this parameter quantifies the degree of certainty with which the mean, calculated from a random sample, estimates the true mean of the population from which the sample was drawn.

$$ES = \frac{\sigma}{\sqrt{n}}$$

σ = dev. population standard

ES depends on the variability in the population and the sample size

Since the distribution of the sample means is a Gaussian distribution with mean μ and standard deviation $\sigma / \sqrt{n}$, we can apply the corresponding properties.

I can then calculate the probability of extracting a sample of size n with a sample mean $\bar{x} = X$

from a population with mean μ and standard deviation σ :

$$Z = \frac{x - \mu}{\sigma / \sqrt{n}}$$

dove:

x : media campionaria

μ : media nella popolazione

$\sigma / \sqrt{n}$: errore standard

Z: deviato normale standardizzata corrispondente ai valori dati per $(x, \mu, \sigma / \sqrt{n})$.

Il valore di probabilità corrispondente al valore Z si legge dalla tabella della distribuzione normale standard.

Summing up

- A statistic calculated on a sample of subjects (sample statistic) constitutes an undistorted estimate of the parameter (the statistic calculated in the population) but the sampling statistic is subject to "sampling error" determined by the random variability of the sampling.
- It is often observed that repeated samples from the same population yield different sample means.
- Each of these averages constitutes an undistorted estimate of the parameter (population mean) but cannot be used as an estimate of the parameter alone, without taking into account the uncertainty due to the error sample book.

We can draw conclusions about the value of the parameter in the starting population from sample data following two paths:

- calculate the confidence interval, or estimate a range of values within which with a predetermined level of probability (generally 95%) it will fall the parameter μ
- perform a hypothesis test with which specific statements about the values of Population parameters can be accepted or rejected

Calculation of the confidence interval: the case of the mean

The sample mean is the best estimate of the mean of population, but that doesn't mean the sample mean is without sampling error: it is easy to observe that samples repeated give different sample means.

The confidence interval of the sample mean is an interval of values around the sample mean, within which it is believed that the parameter under consideration is included with a certain degree of confidence.

The confidence interval is usually referred to a two test code.

Confidence interval for the mean μ of the population with

- known variance σ^2 (of the population)
- sample from a Normal population or sample size > 30 is defined in such a way as to satisfy the following equation

$$p\left(\bar{X} - Z_{\frac{\alpha}{2}} * \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\frac{\alpha}{2}} * \frac{\sigma}{\sqrt{n}}\right) = 1 - p(Z_{\frac{\alpha}{2}})$$

The diagram shows the confidence interval formula. Two arrows originate from the formula: one points from the lower bound $\bar{X} - Z_{\frac{\alpha}{2}} * \frac{\sigma}{\sqrt{n}}$ to a box labeled "Lower fiducial limit", and the other points from the upper bound $\bar{X} + Z_{\frac{\alpha}{2}} * \frac{\sigma}{\sqrt{n}}$ to a box labeled "Upper fiducial limit".

Where

$\bar{X}$ = sample mean

μ = population mean

$\sigma / \sqrt{n}$ = standard error of the mean

$Z_{\frac{\alpha}{2}}$ = value of the standardized normal deviation corresponding to the desired precision. It is calculated by the distribution of the sample means

al 90%
 $\alpha=10\%$

$$\left(\bar{x} - 1,645 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1,645 \frac{\sigma}{\sqrt{n}} \right)$$

Standard Error

al 95% α
 $=5\%$

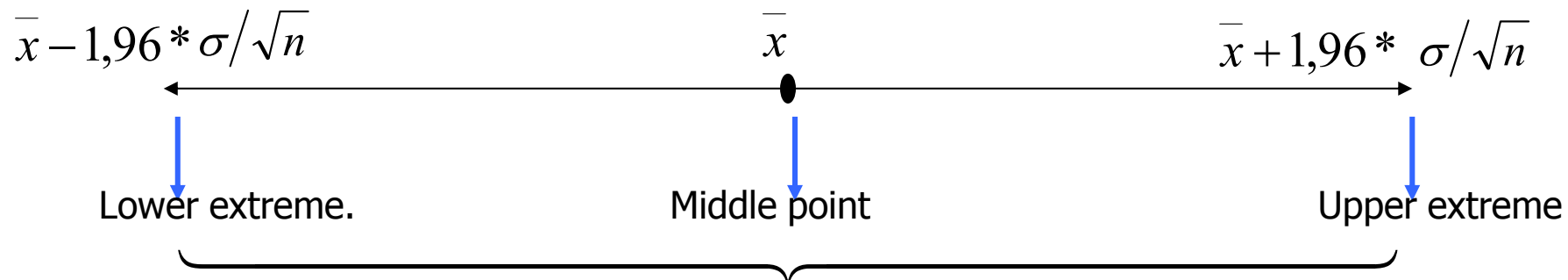
$$\left(\bar{x} - 1,96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1,96 \frac{\sigma}{\sqrt{n}} \right)$$

al 99%
 $\alpha=1\%$

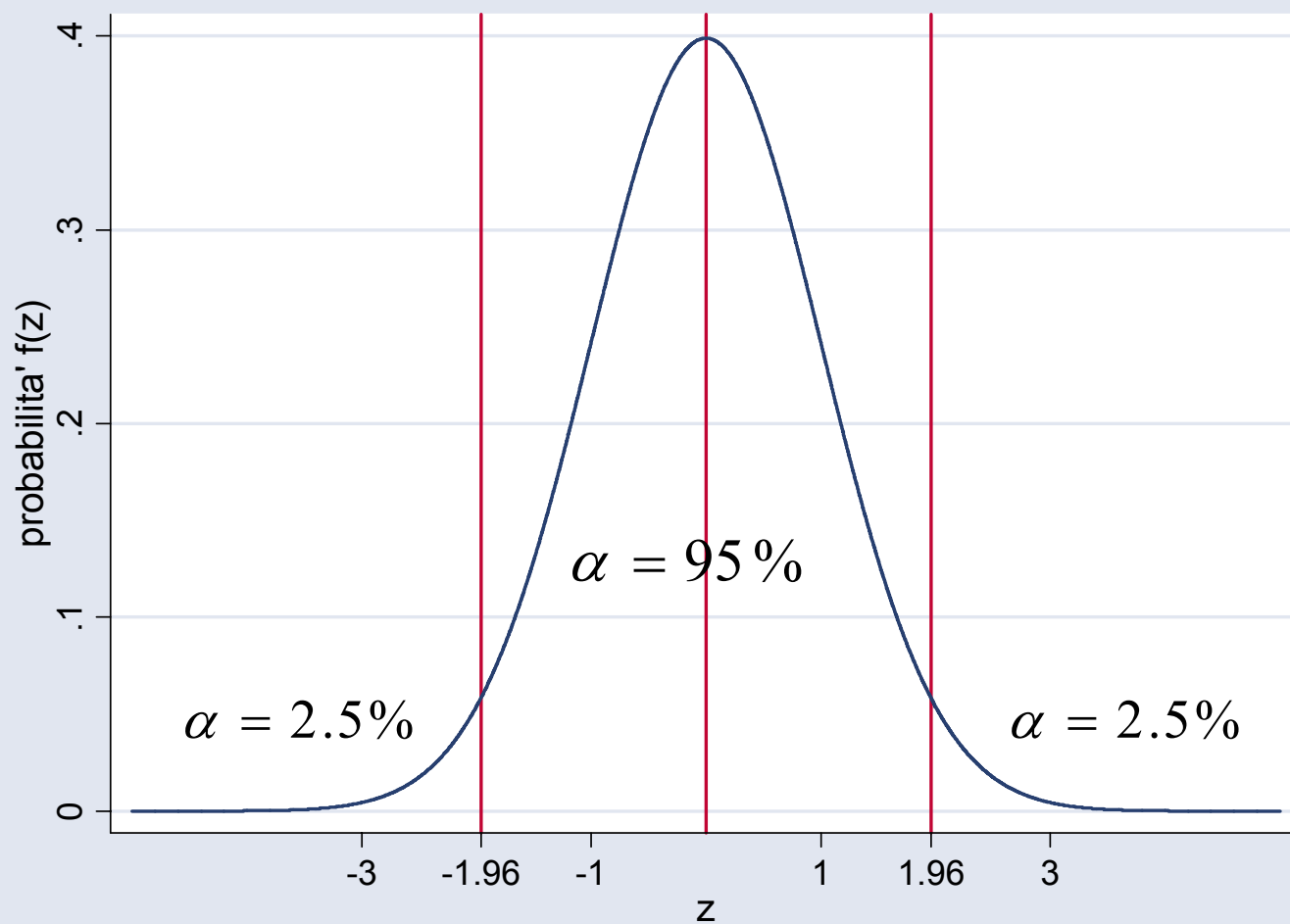
$$\left(\bar{x} - 2,58 \frac{\sigma}{\sqrt{n}}, \bar{x} + 2,58 \frac{\sigma}{\sqrt{n}} \right)$$

Quantili $1 - \alpha/2$ della Normale Standard

Graphical representation of the confidence interval (eg. 95%):



$$\text{Width of the interval} = 2 * (1,96 * \sigma / \sqrt{n})$$





- The most used significance levels (α) are 0.05 and 0.01, which correspond to the confidence coefficients 0.95 and 0.99, often expressed in percentage terms (95% and 99%).

- The width of the interval is determined based on the degree of probability chosen a priori by the researcher: it should be noted that extending this interval to get closer to certainty (100%) has a cost, in how much the inaccuracy of the estimate increases.

We have so far assumed that the variance σ^2 of the population is known but in reality this parameter is rarely known.

How to proceed?

It is necessary to give a precise estimate starting from the sample.

We estimate σ^2 with the sample variance, an undistorted (i.e. unaffected by systematic error) estimator of the population variance:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$



However, the variance of the sample is affected by random variability with respect to the variance of the population, due to sampling.

Even if all individuals in a population were included in the study, they could still be viewed as a sample of one potential biological experience of a larger conceptual population. (Modern Epidemiology II edition, Rothman & Greenland).

Therefore the reference distribution for the inference will no longer be the Normal.

Gosset (who published under the pseudonym of Student) proposed to use a family of distributions with symmetric shape and amplitude depending on the number of observations in the sample: **Student's t functions.**

Confidence interval for the mean μ of the population with

- unknown variance σ^2 (of the population)
- sample from a Normal population or sample size > 30

is defined in such a way that it satisfies the following equation

$$\left(\bar{x} - t_{1 - \frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{1 - \frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}} \right)$$

Lower fiducial limit

Upper fiducial limit

It depends on the sample size!

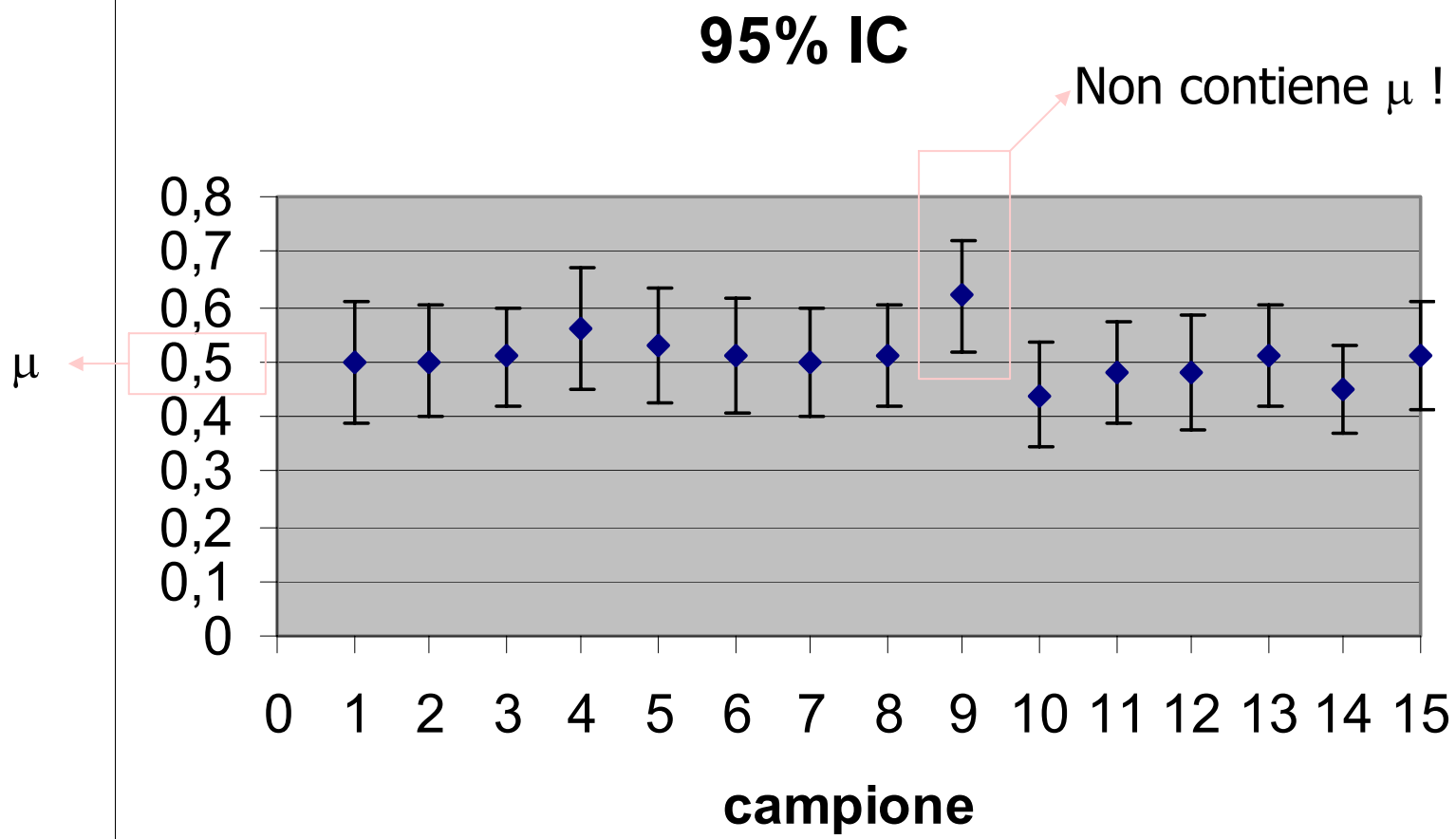
The following meaning must be attributed to the confidence interval for the mean μ :

Given a population with probability distribution characterized by the parameter μ and extracted numerous independent random samples, it is verified that each sample provides a different $\bar{X}$ estimate and therefore the confidence interval is positioned differently with respect to μ .

The value of the unknown parameter (μ , population mean) will be contained in the $1-\alpha\%$ (eg 95% or 99%) of these intervals; therefore 5% of the intervals will not include μ .

In practice, the researcher extracts only one of these samples and therefore observes a single interval to which he attributes a probability $1-\alpha\%$ of containing the true value of μ .

That is to say, the researcher does not know if the 95% CI includes the parameter μ or not, but he knows that the probability that he understands it is 95%.



Exercise

A random sample of 17 women was selected from the population of pregnant women whose fetuses are undergoing postpartum surgery.

The mean hospital stay of this sample was 9.5 days and the standard deviation was 3.6 days.

Calculate the 95% confidence interval for the true population mean and explain what this interval represents.

Reply: $9.5 - 1.96 * 3.6 / \sqrt{17}; \quad 9.5 + 1.96 * 3.6 / \sqrt{17}$

IC95%=7.79-11.21 days

We can say that the true average of hospital stay, with a confidence level of 95%, is between 7.79-11.21 days.

Exercise

It has been stated that the average weight of healthy infants 12 hours after birth is assumed to be 3,400 g. A sample of newborn babies belonging to low-income families provided the following weight values 12 hours after birth:

2721, 3900, 3400, 3720, 3630, 3770, 2930, 2721, 3270, 2180

Calculate the 95% confidence interval for the average weight of children born to low-income families and comment on this interval.

Reply: $3224.2 - 1.96 * 565.8 / \sqrt{10}$; $3224.2 + 1.96 * 565.8 / \sqrt{10}$

IC95%=2873.5- 3574.9 gr

The true weight of children born to low-income families with a 95% confidence level is between 2873.5- 3574.9 g. One of the possible values of the true weight of these children is the same as that of the general population, so the social status does not influence the weight of the newborns.

Hypothesis Test

Statistical test is defined as the tool that allows you to decide about the reliability of the hypothesis (reject or not reject the hypothesis).

In particular, the test is a procedure which, on the basis of sample data and with a certain degree of probability, makes it possible to decide whether it is reasonable to reject the null hypothesis H_0 (and implicitly accept the alternative hypothesis H_1) or if there are insufficient elements for reject it.

The statistical hypothesis is an assumption on the value of a population parameter (eg. The mean μ).

Logical procedure:

1. The researcher formulates a working hypothesis, which constitutes the explanation of a phenomenon or the value of a parameter.

This is not a statistical process, but multiple factors, including:

- study of previous literature
- study of other phenomena related to what we want to investigate
- empirical observations
- ability of the researcher

2. Formulation of the null hypothesis (H₀):

H₀ is the assumption that you want to verify, formulated in such a way that it can be denied by the planned experiment.

For example, if you want to verify the difference between the means of two samples, the null hypothesis will be that there is no difference between the means. So :

$$H_0: \mu_1 = \mu_2$$

3. Formulation of the alternative hypothesis or working hypothesis (H₁):

that can be

$$H_1: \mu_1 \neq \mu_2 \quad \text{or}$$

$$H_1: \mu_1 < \mu_2 \quad \text{or}$$

$$H_1: \mu_1 > \mu_2$$

The alternative hypothesis must be accepted if the null hypothesis is rejected

4. Determination of the significance level of the test (α)

In the hypothesis testing process we apply probabilistic methods and therefore we can find ourselves **rejecting the null hypothesis when it is true.**

In this case we speak of 1st type error or α error.

REFUSAL OF NULL ASSUMPTION (H_0) WHEN IT
IS TRUE $\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true})$

LIKELIHOOD OF DETECTING A DIFFERENCE WHEN IT DOESN'T EXIST

The smaller this probability (α) is, the more unlikely it is that the null hypothesis is true (there is actually a difference or an effect: hypothesis alternative, H_1)

By convention, absolutely arbitrary, the probability is usually considered small if $<5\%$ ($p < 0.05$). If we want to be even more conservative in the defense of H_0 we set $\alpha = 0.01$.

5% is the significance level, indicated with α

5. Measure of the risk of not rejecting the H_0 when this is false

In this case we speak of 2nd type error or β error

I DO NOT REJECT THE NULL HYPOTHESIS (H_0) WHEN IT IS FALSE

$$\beta = P(\text{not reject } H_0 \mid H_0 \text{ is false})$$

PROBABILITY OF NOT DETECTING A DIFFERENCE WHEN IT EXISTS.

Complementary to the β error we have the power of the study:

$$\text{Power} = 1 - \beta$$

Usually the study is designed so that the power is between 80% and 90% ($\beta = 0.2, 0.1$)

By setting β in advance (i.e. error II that we are willing to commit) it is possible to calculate a sample size that will allow us to demonstrate the difference (the effect) if there is.

Significance test	Reality	
	The population mean is μ_0	The average of the population is not μ_0
Not significant (non-rejection H_0)	correct	Type II error β
Significant (refusal H_0)	Type I error α	correct

α (livello di probabilità massima per l'Errore di Tipo I) è la probabilità massima di errore che si intende accettare concludendo che l'ipotesi zero è falsa quando in realtà è vera (errore di I tipo).

Tale probabilità viene solitamente fissata pari a 0,05 (5%) o a 0,01 (1%)

Il valore $(1-\alpha)$ rappresenta il **livello di significatività**

il valore critico (VC) è il valore dell'indice statistico con una probabilità pari ad α

6. Determination of the sample distribution of the test statistic (choice of the appropriate statistical test)

The statistical test is used to calculate the probability associated with the result observed by the experiment.

Under H_0 (= assuming that H_0 is true), the sample distribution of the instrument is determined (test, statistic, test-statistic) which allows to decide between H_0 and H_1

7. The experiment is conducted and the results interpreted

If the probability of obtaining the observed result (given the H_0) is lower than the threshold value defined a priori (that is, if $p < \alpha$), it ends with the rejection of the null hypothesis.

Otherwise it ends with the non-rejection of the null hypothesis.

When a $p > 0.05$ is obtained (i.e. an insignificant result) it does not mean that H_0 is true but only that there is not enough evidence to reject it

**Confrontando il valore
di P con α :**

se $P > \alpha$

allora accetto l'ipotesi zero

se $P \leq \alpha$

allora rifiuto l'ipotesi zero



How do we arrive at probability P?

By calculating a test statistic

Z for comparison with the general population when the variance is known

t for comparison with the general population when variance is unknown or for comparison between two samples

$$\frac{\text{observed value} - \text{expected value}}{\text{standard error of the observed value}}$$

to which a certain value of p will correspond

Meaning of p

It is the probability of having observed our data (or even + extreme data) in the event that the null hypothesis (H_0) were true!

- The choice of $P < 0.05$ as the cut-off for rejecting H_0 is absolutely arbitrary!
- When you get a $P > 0.05$ (i.e. an insignificant result) not it means that H_0 is true but only that there is not enough evidence for reject it

Let us dwell on the P

It is always necessary to indicate the exact value of P (not only if it is $\leq$ or $\geq$ 0.05) and then interpret

- $P = 0.045$ (significant) and $P = 0.055$ (not significant) must not automatically lead to opposite decisions !!!!
- A study on a large sample may show small ones differences, not very relevant but statistically very significant while a study with a small sample may not highlight important differences!
- It is therefore always better to associate the estimate of the effect and its IC with the P value.

There are many tests available:

The appropriate test is chosen in relation to the type of samples to be compared (independent, for paired data) and according to the variable on which the comparison is to be carried out.

INDEPENDENT SAMPLES: the data are provided by **different subjects** (e.g. Two different groups, one subjected to treatment and the other not)

EMPLOYEES SAMPLES (PAIRED DATA): the data are provided by the **same subjects** (eg the same group analyzed before and after the treatment).

- **PARAMETRIC TESTS** for quantitative variables with known distribution.
- **NON PARAMETRIC TESTS** for qualitative or quantitative variables with unknown distribution

ONE OR TWO-Tailed TEST?

One-sided test: A critical value is found which leaves in one single tail a probability equal to p .

In rare cases it is reasonable to think that the real difference can occur in only one direction (new drug better than the old one), while a difference in the opposite direction must be due to chance: a one-tailed P is calculated (unilateral test)

Two-tailed test: two critical values are identified which respectively leave a probability equal to $p / 2$ on the left and on the right: if the test statistic is within this region, I do not reject the H_0 .

When we try to prove a difference, it can happen in **both directions** (the test is bilateral)

For the 5% significance level

Z two-tailed = 1.96

Z one-side = 1.64 (less conservative test !!)

TEST Z

Parametric test. Valid provided the variable is normally distributed and the population variance is known

TEST THE HYPOTHESIS FOR A POPULATION WITH AVERAGE μ : σ^2 KNOWN

$$z = \frac{\text{media campione} - \text{media popolazione}}{\text{errore standard}}$$

$$z = \frac{\text{differenza osservata}}{\text{differenza dovuta alla casualità}}$$

I use the population variance to calculate the standard error

Exercise

Is the mean cholesterol level in the hypertensive population equal to the mean cholesterol level in the normal population at a significance level of 5%?

Let's imagine we know the mean serum cholesterol level (μ) of the male population 20-74 years of age in the USA, which is equal to

$$\mu = 211 \text{ mg/100 ml}; \sigma = 75 \text{ mg/dl}$$

We extract a sample of 25 hypertensive subjects and measure the cholesterol level; the sample mean is

$$X_{\text{medio}} = 228 \text{ mg/100ml.}$$

Is the mean serum cholesterol level in the hypertensive population (μ_0) the same as that in the general population (μ)? Is the observed difference due to chance?

To answer this question, a hypothesis test can be used to make inferences on the cholesterol value in the hypertensive population from the data observed in a sample.

Let's build the hypotheses

A theoretical hypothesis is initially posed:

"The mean in the hypertensive population (μ_0) is the same as that of the population general (μ)? "

The null hypothesis (H_0) assumes that there is no difference between the two populations.

$$H_0 : \mu = \mu_0 \quad (\mu = 211\text{mg} / 100\text{ml})$$

Instead, the alternative hypothesis (H_1) affirms the opposite of the null hypothesis: that is, that the mean cholesterol in hypertensive is different from that of the population normal.

$$H_a : \mu \neq \mu_0 \quad (\mu \neq 211 \text{ mg} / 100 \text{ ml})$$

We need to calculate the value of z and look on the tables how probability is associated with this value.

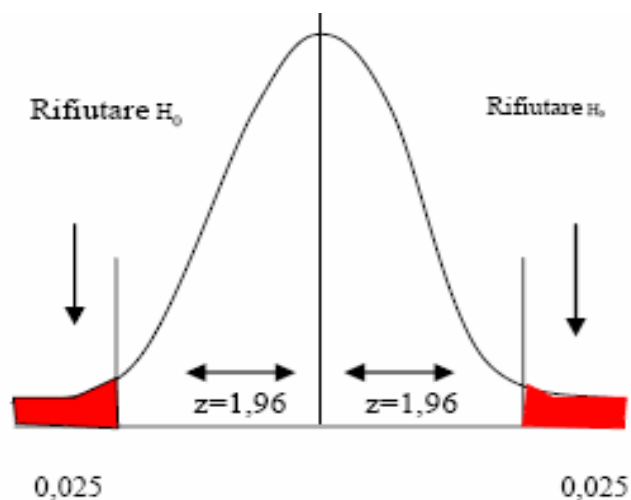
$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$\bar{x}$	sample mean
μ_0	population mean
$\sigma / \sqrt{n}$	standard error
n	sample size

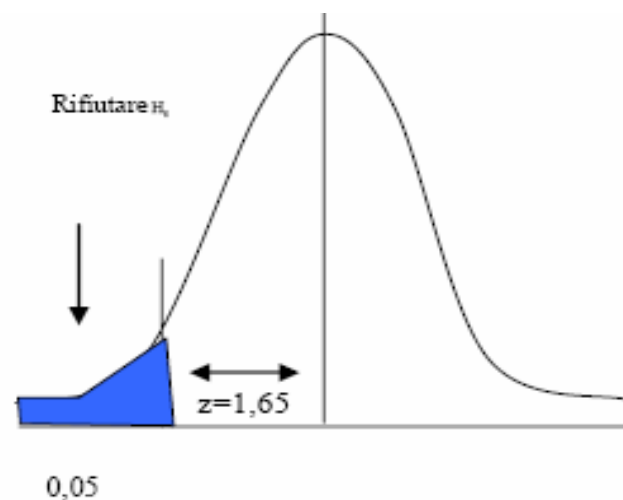
$$Z = (228 - 211) / (75 / \sqrt{125}) =$$

$$Z = 17 / (75 / 11.18) = 17 / 6.7 = 2.54$$

$$P(z \geq 2.54) = 0.0012$$



$\alpha=0,05$



$\alpha=0,05$

Since $p = 0.0012$ is less than the significance level $\alpha = 0.05$ we conclude that the test is statistically significant: the cholesterol value in hypertensive people is different from that of the normal population at a significance level of 5%.

The probability that a sample extracted from the population with mean $\mu = 211$ and $\sigma = 75$ has an average equal to that found in our sample, that is $\bar{x} = 228$ is equal to 0.0012

Conclusion: the probability is less than 5%, so I can reject the null hypothesis, so there is a difference between the average cholesterol in hypertensive subjects compared to the general population.

Exercise

The population of non-smoking women aged 3 to 74 who participated in the National Health Interview Survey has a mean hemoglobin value of $\mu = 13.3$ g / 100ml, and a standard deviation of 1.12.

A sample of 80 female smokers reported a mean hemoglobin value of 11.

Is the mean hemoglobin value of women smokers (μ_0) equal to that of non-smokers at a significance level of 5%?

We build the hypotheses:

Null hypothesis:

$$\mu = \mu_0$$

$$\mu = 13.3, \quad \sigma = 1.12$$

alternative hypothesis:

$$\mu \neq \mu_a$$

$$n = 80, \quad X_{\text{medio}} = 11$$

Is the mean hemoglobin value of women smokers (μ_0) equal to that of non-smokers (μ) at a significance level of 5%?

$$Z = (x_{\text{medio}} - \mu) / (\sigma / \sqrt{N}) =$$

$$Z = -2.3 / (1.12 / 8.944) =$$

$$Z = (11 - 13.3) / (1.12 / \sqrt{80}) =$$

$$Z = -2.3 / 0.125 = 2.17$$

$$P(z \geq 2.17) = 0.04$$

Conclusion: The p-value, (the probability associated with the result of this study) is less than 5%, so I can reject the null hypothesis, and conclude that there is a difference between the mean hemoglobin in smokers versus women. non-smokers.